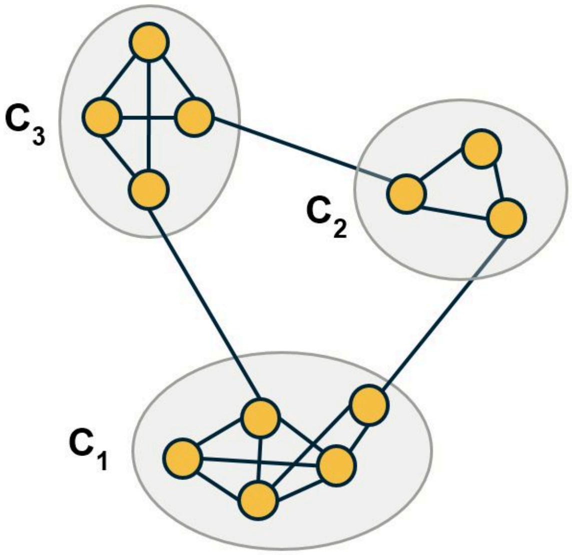
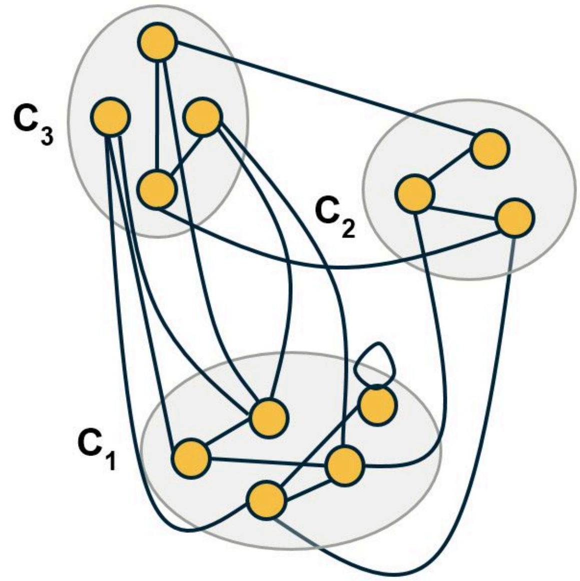
L7: Modularity Metric



Given network and assignment of nodes to three hypothetical communities



A randomized network where each node has the same degree as in the given network

All approaches for community detection we have seen so far cannot answer the following two questions:

Which level of the dendrogram, or more generally, which partition of the nodes into a set of communities is the best?

And, are these communities statistically significant?

The modularity metric gives us a way to answer these questions. The idea is that randomly wired networks should not have communities -- and so a given community structure is statistically significant if the number of internal edges within the given communities is much larger than the number of internal edges if the network was randomly rewired *(but preserving the degree of each node).*

In more detail: suppose we are given a network with *N* nodes and *L* edges – for now, let us assume that the network is undirected and unweighted.

We denote the degree of node *i* as k_i.

Additionally, we are given partitioning of all nodes to a set of *C* hypothetical communities $C_1, C_2, \ldots C_C$.

Our goal is to evaluate this community structure – how good it is and whether it is statistically significant.

Let *A* be the network adjacency matrix.

The number of internal edges between all nodes of community C_c is

$\frac{1}{2}\sum_{(i,j)\in C_c} A_{i,j}$

On the other hand, if the connections between nodes are randomly made, the expected number of internal edges between all nodes of that community is,

$\frac{1}{2}\sum_{(i,j)\in C_i} \frac{k_i k_j}{2L}$

because node i has k_i stubs and the probability that any of those stubs connect to a stub of node *j* is $\frac{k_j}{2L}$

So, we can define the modularity metric based on the difference between the actual number of internal edges and the expected number of internal edges, across all C communities:

$$M = \frac{1}{2L} \sum_{c=1}^{C} \sum_{(i,j) \in C_c} (A_{i,j} - \frac{k_i k_j}{2L})$$

We divided by the total number of edges L, so that M is always less than 1.

Note that the modularity metric does not directly penalize any external edges, i.e., there is no term in this equation that decreases the modularity for every external edge between two communities. The more external edges exist however, the lower is the sum of the internal edges, while the sum of the expected number of random edges remains constant. In other words, the modularity metric indirectly selects community assignments that have more internal edges and fewer external edges.

How can we use this metric to select between different community structures? It is simple: select the set of communities that has the larger modularity value.

And how can we know if a given community structure is statistically significant? A simple way to answer this question is by comparing the modularity of a network with 0, which is the value we would expect from a randomly wired network.

Alternatively, we can generate an ensemble of random networks using degree-preserving randomization and estimate the distribution of modularity values in that ensemble. We can then use a one-sided hypothesis test to evaluate whether the modularity of the original network is larger than that distribution, for a given statistical significance level.